Journal of Environmental Management 377 (2025) 124600



Contents lists available at ScienceDirect

Journal of Environmental Management

journal homepage: www.elsevier.com/locate/jenvman



Research article



Mapping the probability of forest fire hazard across the European Alps under climate change scenarios

Kilian Gerberding [a], Uta Schirpke [b,*]

[a] Faculty of Environment and Natural Resources, University of Freiburg, Tennenbacher Str. 4, Freiburg, 79106, Germany
[b] Institute for Alpine Environment, Eurac Research, Drususallee 1, Bozen/Bolzano, 39100, Italy

ARTICLE INFO

Keywords:
Global warming
Shared socio-economic pathways (SSPs)
Mountain forest
Wildfire
Geostatistical model
Geographically weighted regression

ABSTRACT

Forest fires are increasing in frequency and intensity worldwide due to the anthropogenic climate change, threatening people's lives and causing huge economic and environmental damages. Recent forest fire events suggest that forest fires are also an urgent issue in the European Alps, but studies assessing the forest fire hazard under future climate scenarios are still rare. Thus, this study aims to analyse the impacts of climate change on the probability of forest fire hazard across the European Alps and surrounding areas. In specific, we (1) explain the current forest fire hazard based on a set of environmental and anthropogenic parameters, and (2) map the forest fire hazard under current and future conditions across the study area using geographically weighted regression. Our results suggest that the fire hazard mainly depends on the frequency of lightning strikes, the annual mean temperature, and the precipitation seasonality. Overall, our results indicate a future increase in forest fire hazard, which is already significant under the SSP126 (+15.5%), while highest increases occur under the SSP370 (30.6%) and the SSP585 (35.4%). However, while the impacts are less pronounced in already fire-prone regions in the southwestern regions in France, the probability of forest fire hazard will greatly increase in the Northern and Eastern regions. Our findings emphasize the urgent need to address these climate-related challenges by decision-making and management through fire-smart forest management. Nevertheless, further efforts are needed to overcome current limitations related to data availability and uncertainties in future scenarios.

1. Introduction

Fires are a major natural disturbance of forest ecosystems, influencing ecological processes, biodiversity, forest structures, and landscape patterns. Depending on the fire regime, forest fires can be a threat to people's lives and cause economic and environmental damages (Bowman et al., 2020). Large intensive fires have huge impacts on the physical health, as the exposure to smoke is associated with an increased risk of respiratory cardiovascular diseases, among others (Johnston et al., 2012; Liu et al., 2015). Experiencing the threat to personal safety and the loss of properties and belongings can have long-lasting consequences on mental health, causing depression, posttraumatic stress disorder, and anxiety (Gao et al., 2023; To et al., 2021). Moreover, forest fires generate considerable economic losses due to damaged infrastructure and settlements, health costs, and the disruption of supply chains (Wang et al., 2021). Depending on characteristics of fires (i.e., size, type, and intensity), there are numerous environmental impacts and related ecosystem services, including carbon emissions, soil degradation, desertification, loss of forests, and biodiversity decline (Agbeshie et al., 2022; Armenteras et al., 2021; Jolly et al., 2022; Phillips et al., 2022). Furthermore, forest fires in mountainous regions endanger protection forests (Stritih et al., 2024), leading to a higher vulnerability to natural hazards like avalanches and landslides, and they reduce cultural ecosystem services such as recreational and aesthetic experiences (Iglesias et al., 2022; Lecina-diaz et al., 2023; Spadoni et al., 2023).

The occurrence of forest fires is associated with various environmental and anthropogenic factors such as weather conditions, vegetation characteristics, and human activities (Gao et al., 2024; Sevinc et al., 2020). Regions characterized by dry climatic conditions have always been prone to forest fires, located, for example, around the Mediterranean in Europe, in western North America, and Australia (Andela et al., 2019; Jones et al., 2022). However, it is expected that human-induced climate change increases the forest fire activity globally, also in regions that have been less affected by forest fires in the past (Bowman et al., 2020; Dupuy et al., 2020; Resco de Dios et al., 2021). It is therefore

* Corresponding author.
E-mail address: uta.schirpke@eurac.edu (U. Schirpke).

https://doi.org/10.1016/j.jenvman.2025.124600
Received 22 October 2024; Received in revised form 27 January 2025; Accepted 15 February 2025
Available online 22 February 2025
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likely that mountainous regions in Europe, such as the European Alps, will be increasingly affected by forest fires for several reasons. First, they are characterized by a high share of forest, potentially providing a huge amount of fuels accumulated, as current climate conditions are too wet to sustain large fires (Resco de Dios et al., 2021; Rigo et al., 2017). In addition, the abandonment of mountain grassland, together with an upshift of the treeline, leads to an expansion of the forest area (Schirpke et al., 2023; Tasser et al., 2017). Climate predictions indicate rapid warming as well as significant changes in seasonal precipitation patterns causing more severe droughts in mountain regions (Kotlarski et al., 2023; Pepin et al., 2022). Moreover, forest fires in mountainous terrain are often difficult to control due to reduced accessibility (Wastl et al., 2013) and faster fire spread on steeper slopes (Janizadeh et al., 2023).

To support decision-makers and forest managers, a context-specific understanding of underlying drivers is needed to develop effective strategies in reducing potential future hazard (Berčák et al., 2023). Moreover, spatially explicit information that reveal changes in spatial patterns can support identifying priority areas for hazard management (Stritih et al., 2024). Despite a growing body of literature modelling the forest fire hazard in different regions across the world (Babu et al., 2023; Bountzouklis et al., 2023; Tran et al., 2024), studies rarely predicted the forest fire hazard under future climate scenarios (Grünig et al., 2023; Shao et al., 2023; Varela et al., 2019). Only a few studies have focused on European mountain regions, for example, addressing the role of fuel moisture content in the Pyrenees (Resco de Dios et al., 2021), assessing the effects of forest management in the French Alps (Jourdan et al., 2021), evaluating the influence of forest dynamics and climate on fire activity in the Swiss Alps (Schumacher and Bugmann, 2006), and predicting the future forest fire danger based on the Canadian Forest Fire Weather Index in French mediterranean regions (Varela et al., 2019), in the Bavarian Alps (Miller et al., 2024), and across the entire European Alps (Cane et al., 2013). Most of these studies provide insights useful for management and planning at the local or regional level, while studies covering larger study areas are still needed, having higher potential to facilitate strategies and policies at multiple governance scales (Koutsias et al., 2016; Montiel-Molina, 2013).

To support decision-making and management in identifying and addressing potential threats to people and mountain forests, this study aims to analyse the effects of climate change on the forest fire hazard across the European Alps and surrounding areas. We hypothesize that increasing temperatures and reduced precipitation increase the pressure on regions that have been less affected by forest fires up to now, while remaining high in already fire-prone regions. Our specific objectives are (1) to explain the current forest fire hazard based on a set of environmental and anthropogenic parameters, and (2) assess effects of climate change on the future forest fire hazard by mapping the probability of forest fire hazard under current and future conditions across the study area. We use statistical and spatially explicit modelling approaches to identify drivers of forest fires as well as to detect changes in spatial patterns in the study area. Our findings provide useful information to decision-makers and forest managers, supporting them to scope with climate change-induced challenges in mountain regions.

2. Material and methods

2.1. Study area

The 'Alpine Space Programme cooperation area' (hereinafter referred to as ASP) extends over an area of 391,000 km^2 and includes Austria, Switzerland, Liechtenstein, Slovenia, as well as several regions of France, Germany, and Italy (Fig. 1). About 70 million people live in the ASP, most of them (81%) in the lowlands in highly urbanized areas (Eurostat, 2021). Moreover, it is an important tourist destination with about 6.8 billion overnights per year. The ASP is centred around the European Alps, comprising mountainous areas (53%) up to 4809 m above sea level (mean elevation 1279 m above sea level) as well as lowland areas (47%) that are located on average at 340 m above sea level (Schirpke, 2022). The high variation in topography results in great climatic differences. While the northern and southeastern regions are characterized by a humid climate with 1500 to 2700 mm precipitation per year, the southern and western regions are dryer with an annual precipitation of 500–950 mm. Mean annual temperatures vary between −6 °C at high altitudes to 14 °C in the southern lowlands. These climatic differences are also reflected in the predominant land use/cover types. The mountainous areas are dominated by forest (49%), grassland (20%), and high-mountain ecosystems with little or no vegetation (13%), whereas the lowlands are characterized by annual and permanent crops (50%), forest (25%), and built-up areas (10%) (Schirpke, 2022). While the Alpine region has undergone several periods of forest fire regimes, since the late 1990s, it has primarily been characterized by periodic

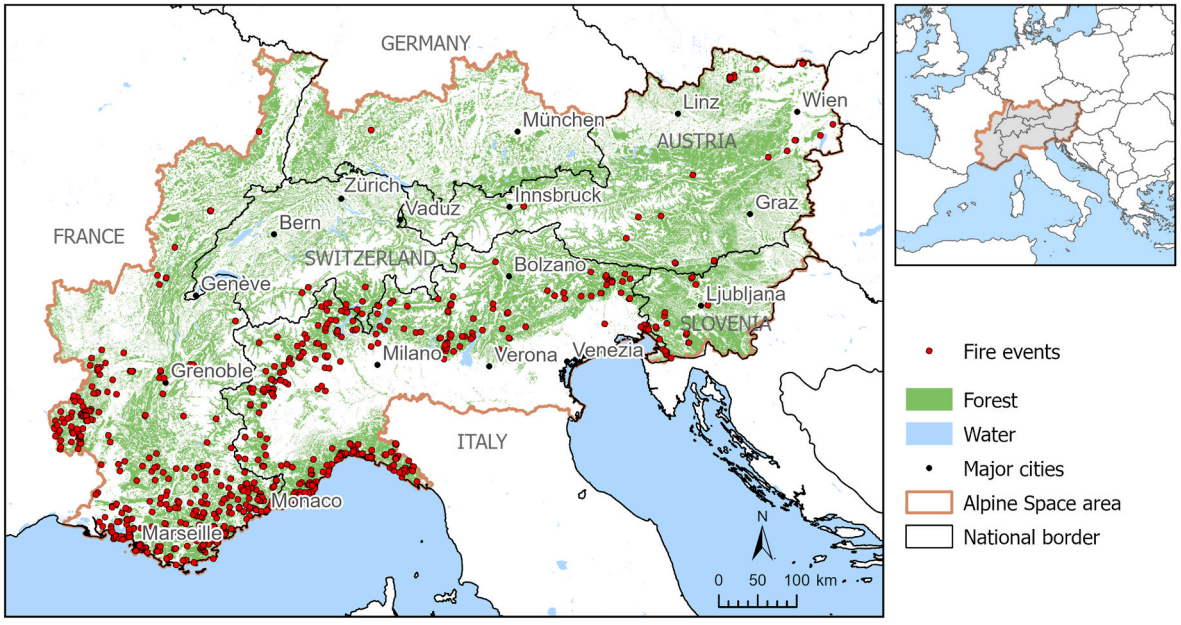


Fig. 1. Location of the study area - the Alpine Space Programme cooperation area (ASP) - and location of past forest fire events (2000–2023).

autumn-winter and early-spring surface fires, along with a growing proportion of lightning-induced summer fires. With the increasing frequency of weather conditions conducive to fires, combined with the lack of forest management leading to the build-up of fire fuel, larger and more intense forest fires have become more common (Valese et al., 2014).

2.2. Methodological approach

To analyse the effects of climate change on the forest fire hazard, we applied four main analysis steps (Fig. 2). To predict the occurrence of forest fires, geostatistical modelling approaches are widely applied, using empirical data to explain past fire events by conditioning factors, including environmental and anthropogenic variables, through statistical and data mining methods (Babu et al., 2023; Iban and Sekertekin, 2022; Jaung and Carrasco, 2021; Tran et al., 2024). In a first step, we therefore created a dataset containing locations of past forest fire events as well as randomly selected locations where no fire has been occurred over the last two decades (section 2.3). The second step comprised the selection and calculation of a set of predictor variables (section 2.4) that covered various potential natural and anthropogenic causes of forest fires (Bountzouklis et al., 2023; Sevinc et al., 2020). In a third step, we related these predictor variables to the dataset containing fire locations and non-fire locations through a binary logistic regression model to explain the occurrence of forest fires (section 2.5). Using Geographically Weighted Regression (GWR), we mapped the forest fire hazard under current climatic conditions across the entire study area (section 2.5). The fourth step included the mapping of future probability of forest fire hazard under three climate change scenarios and the assessment of the effects of climate change by spatially overlaying the different maps (section 2.6).

2.3. Preparing the forest fire database

To model fire hazard across the ASP, we used the European Forest Fire Information System (EFFIS; https://forest-fire.emergency.copernicus.eu/), providing harmonised data of burnt areas in the European Union. We first downloaded the perimeters of all burnt areas between 2000 and 2023 located in our study area. In a next step, we removed all burnt areas that included only other land cover types than forest (i.e., CLC classes 311, 312, 313, 323, 324) by overlying the burnt areas with the CLC types, resulting in 772 fire locations distributed across the study area. To model the probability of forest fire hazard through binary regression modelling, we attributed these fire locations a value of 1, while we also created non-fire locations in forest across the study area, assigning them a value of 0 (Fig. S1a). We thereby accounted for spatial patterns in the past forest fire events: First, to ensure that non-fire locations had not been already affected by forest fires in the past and given the spatial clustering of the fire-locations, we restricted the placement of the non-fire locations as follows. We calculated the distance between all burnt areas included in the EFFIS database using Delaunay triangulation, which includes only features as neighbours that share edges or corners (Scott and Janikas, 2010). This resulted in an average median distance of 17.4 km as buffer around the fire locations. After creating a buffer around the fire locations, we removed these buffer zones from the total forest area across the study area to obtain forest areas for placing the non-fire locations. Second, considering that the spatial extent between regions affected by forest fires and those without forest fire events differed, which derived the number of non-fire locations based on the ratio between these two regions, i.e., 1:2.25 (fire to non-fire regions). Thus, we created 1739 random non-fire points across the non-fire region, resulting in a total sample of 2511 fire/non-fire locations (Fig. S1). Finally, we added an area with a radius of 325 m around the non-fire points to obtain non-fire locations, which corresponds to the median of the area (0.33 km^2) of burnt areas included in the EFFIS database in the ASP.

2.4. Calculating the predictor variables

Various environmental and anthropogenic factors influence the occurrence of forest fires. To account for the diversity of drivers, we considered 24 variables related to forest fire hazard, including environmental and anthropogenic variables (Table 1). The selection of these variables was based on current literature (e.g., Babu et al., 2023; Bountzouklis et al., 2023; Carrasco et al., 2021; Iban and Sekertekin, 2022; Qadir et al., 2021; Tran et al., 2024), but it was also influenced by the availability of data. For example, it was not possible to include relevant variables such as information on understory fuels (Sánchez-Pinillos et al., 2021), fuel properties of litter and dead wood (Hanes et al., 2021; Morgan Varner et al., 2015), as well as on stand characteristics such as age, density, and management (Rigo et al., 2017). All potential predictor variables (Table 1) were mapped at a spatial resolution of 100 m across the study area (for details see subsections below).

2.4.1. Environmental variables

In mountain regions, topography is a decisive factor for climate, weather, vegetation, and land-use/cover patterns. Thus, elevation plays a key role in the general occurrence of forest fires (Bountzouklis et al., 2023; Qadir et al., 2021), while slope and aspect lead to great differences in local temperature and soil moisture, which affects the forest fire hazard through higher probability of ignition and faster fire spread (Janizadeh et al., 2023). For each fire/non-fire location, we therefore derived the mean elevation, mean slope, and mean aspect from a Digital Elevation Model (DEM; https://spacedata.copernicus.eu/de/collections/copernicus-digital-elevation-model). Moreover, we calculated the Topographic Wetness Index (TWI), indicating hydrological behaviour based on terrain characteristics (Beven and Kirkby, 1979). Since soil moisture is an important determinant for forest fires, influencing ignition conditions and flammability (O et al., 2020), we employed the TWI as a proxy for local soil moisture conditions (Kopecký et al., 2021).

Weather and climatic conditions are one of the main factors for forest

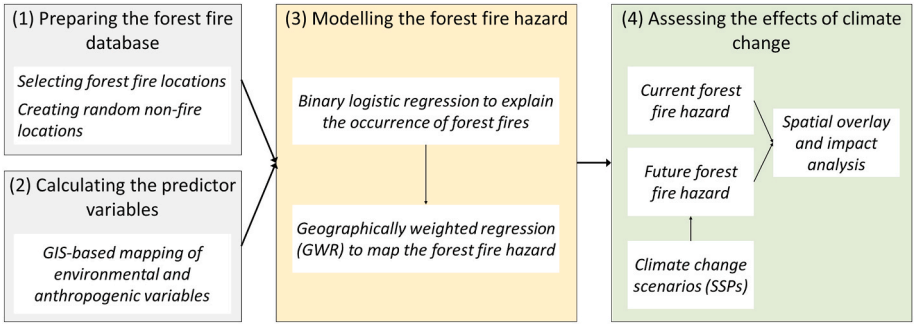


Fig. 2. Conceptual steps for modelling and assessing the effects of climate change on the forest fire hazard.

Table 1
Potential predictor variables for mapping the forest fire hazard.

Group	Variable	Description	Unit	Data sources
Environmental	Elevation	Mean elevation	m above sea level	DEM[a]
	Slope	Mean slope	degrees	DEM[a]
	Aspect	Mean aspect	degrees	DEM[a]
	TWI	Topographical wetness index	index	DEM[a]
	Temperature	Mean annual temperature	°C	CHELSA[b]
	Precipitation	Mean annual precipitation	mm	CHELSA[b]
	Precipitation Seasonality	Standard deviation of the monthly precipitation estimates expressed as a percentage of the annual mean precipitation	mm	CHELSA[b]
	Lightnings	Mean density of lightnings strikes	n km^{-2}	Met Office[c]
	TCD	Tree cover density	%	CLMS[d]
Anthropogenic	Road distance	Euclidean distance from the nearest paved road	m	OSM[e]
	Railway distance	Euclidean distance from the nearest railway track	m	OSM[e]
	Road time	Walking time from the nearest paved road	minutes	OSM[e]
	Settlement time	Travel time from the nearest settlement (CLC classes 111 and 112)	minutes	OSM[e], CLC[f]
	Town time	Travel time from the nearest town (i.e., important urban centre, between a village and a city in size)	minutes	OSM[e]
	City time	Travel time from the nearest city (i.e., largest urban settlement within the territory, including national, regional, and provincial capitals)	minutes	OSM[e]
	Agriculture distance	Euclidean distance from agricultural areas (CLC classes 211–244)	m	CLC[f]
	River distance	Euclidean distance from the nearest river	m	OSM[e]
	Water distance	Euclidean distance from water (CLC classes 511 and 512)	m	CLC[f]
	Population	Population density in proximity to the fire locations (buffers of 1, 3, 5, and 10 km)	n km^{-2}	Eurostat[g]
	Visitation Flickr	Density of photos from Flickr (i.e., photo-user days)	n km^{-2}	Flickr[h]
	Visitation GBIF	Density of iNaturalist research-grade observations	n km^{-2}	GBIF[i]

[a] DEM: Digital Elevation Model (https://spacedata.copernicus.eu/de/collections/copernicus-digital-elevation-model).
[b] CHELSA: CHELSA V2.1 Climatologies (https://chelsa-climate.org/).
[c] Met Office (https://www.metoffice.gov.uk/).
[d] CLMS: Copernicus Land Monitoring Service (https://land.copernicus.eu/en/products/high-resolution-layer-tree-cover-density/tree-cover-density-2018).
[e] OSM: OpenStreetMap (https://www.openstreetmap.org/).
[f] CLC: Corine Land Cover 2018 (https://land.copernicus.eu/pan-european/corine-land-cover/clc2018).
[g] Eurostat: Eurostat census grid 2021 (https://ec.europa.eu/eurostat/web/gisco/geodata/grids).
[h] Flickr: Photo-sharing website Flickr (https://www.flickr.com/).
[i] GBIF: Global Biodiversity Information Facility (https://www.gbif.org/dataset/50c9509d-22c7-4a22-a47d-8c48425ef4a7).

fires affecting the ignition and behaviour of forest fires (Estes et al., 2017; Gao et al., 2024). We used the mean annual temperature, the mean annual precipitation sum, and the precipitation seasonality (1981–2010), which were extracted from the CHELSA V2.1 data repository (Karger et al., 2017, 2021). Furthermore, lightning strikes have been identified as the most important natural causes for forest fires, depending on both the frequency of lightning strikes as well as environmental conditions (Castedo-Dorado et al., 2011; Müller et al., 2020; Sevinc et al., 2020). Although the occurrence of lightning fires is mainly influenced by dry flash, we had to use the density of lightning strikes as predictor variable due to limited data availability across the study area. To calculate the mean density of lightning strikesbased on the location system ATDnet, we grouped every detection of the system (fix) (2009–2023) occurring within a radius of 20 km and 1 s of another one into one flash (Anderson and Klugmann, 2014; Drüe et al., 2007). However, it needs to be considered that the ATDnet system cannot differentiate between cloud-to-ground lightning strikes or powerful cloud pulses, which may result in an overestimation of actual density of fire-igniting lightning strikes (Enno et al., 2020).

The characteristics of tree species and the availability of fuels influence the severity of forest fires (Popović et al., 2021; Walker et al., 2020). We therefore included. Second, t the tree cover density (TCD; EEA, 2020), which represents the proportional tree crown coverage in each pixel with values ranging from 0 to 100%, to account for the distribution of potential fuel in the study area.

2.4.2. *Anthropogenic variables*

The occurrence of forest fires is strongly correlated with human activities through accidental causes, arson, and people's negligence (Bountzouklis et al., 2023; Sevinc et al., 2020). Most studies use indicators such as distance to roads and settlements to indicate the level of human influence (Babu et al., 2023; Iban and Sekertekin, 2022; Tran et al., 2024). However, these indicators do not sufficiently reflect the level of accessibility in mountainous landscapes as well as the presence of people. In addition to the Euclidean distance to roads as well as railways, we therefore calculated the distance to roads in terms of walking time. Both indicators were based on paved roads included in the OpenStreetMap (OSM; https://www.openstreetmap.org/). The walking time was derived according to Tobler's hiking function that indicates the walking velocity as a function of slope (Tobler, 1993). The distance from settlements, towns, and cities was expressed as travel time by car (on paved roads) and by foot (on roads and paths not accessible by car, as well as off-road). From the road network provided by OSM, we derived the maximum speed of the roads. Missing data were integrated by assigning each type of road a mean travelling velocity, and an average off-road velocity of 1 km per hour was assumed. Settlement areas were extracted from the CORINE land cover database (European Environment Agency, 2020) and the travel time was estimated using the cost distance algorithm as implemented in ArcGIS Pro 3.2 (ESRI, Redlands, CA, USA).

A further factor for the occurrence of forest fires is related to the land use, especially to agricultural practices such as the burning of agricultural and forest debris (Carrasco et al., 2021). We therefore derived the distance of agricultural areas extracted from the Corine land cover database (European Environment Agency, 2020) to the fire/non-fire locations. Moreover, the distance from water can affect the occurrence

of forest fires, potentially mitigation their ignition and spread due to higher soil moisture as well as the availability of water for extinguishing the fire (Tran et al., 2024). We therefore included two indicators related to the distance of water, calculating the Euclidean distance from (1) rivers provided by OSM, and (2) large water courses and water bodies included in the CORINE land cover database (European Environment Agency, 2020).

To capture the number of people living in proximity to the burnt areas and non-fire locations, we derived the population density from the Eurostat census grid 2021 (https://ec.europa.eu/eurostat/web/gisco/geodata/grids), applying buffer zones of 1, 3, 5, and 10 km radius. To also detect differences in the presence of people carrying out (recreational) activities in the landscape, we used mean visitation rates that were derived from social media data, using (1) the photo-sharing website Flickr (https://www.flickr.com/) and (2) research-grade observations from the iNaturalists database (https://www.gbif.org/dataset/50c9509d-22c7-4a22-a47d-8c48425ef4a7). Flickr hosts a large repository of photographs related to recreational activities (Schirpke et al., 2018). We used the metadata of all public photos in the study area taken at the street level to assure best spatial accuracy. We focused on photos taken in (semi-)natural land cover types, excluding urban and artificial areas (i.e., CORINE land cover classes 111–133). In contrast, the scope of the citizen science project iNaturalist differs from Flickr, as it allows users to identify and share biodiversity information uploading pictures of the species as well as their location (Aristeidou et al., 2021). We downloaded the metadata of observations with spatial accuracy of over 20 m during 2020–2023. For each dataset, we calculated the point density of photographs taken/observations made by one user per day in a specific location applying a moving window with a search radius of 1.5 km (Schirpke et al., 2018). To reduce uncertainties related to the social media data (i.e., low photo density in some land cover types, uncertainties in the location of photographs, spatial aggregation), we used a 300 m radius buffer to calculate the mean visitation rates for each location.

2.5. Modelling the forest fire hazard

To select indicators containing enough independent information and avoid issues of multicollinearity, we first determined the Pearson's R and excluded variables with $R > 0.7$ (Table S1), indicating that they had more than 50% of the information in common. We also excluded variables with a variance inflation factor (VIF) > 5 indicated during regression analysis. The remaining 16 predictor variables were used to explain the occurrence of forest fires through binary logistic regression the with ArcGIS Pro 3.2 (ESRI, Redlands, CA, USA). We first applied generalized linear regression (GLR) with the binary logistic model. As the residuals of the GLR model were spatially clustered indicating that the occurrence of forest fires depended on the local/regional context, we applied GWR to map the current forest fire hazard across the entire study area. GWR is suitable to capture spatial variability in the relationships, as it establishes separate models only based on the data from the neighbourhood (Fotheringham et al., 2009). In the GWR, we included only eight variables resulting significant from the binary logistic regression (Fig. S2), i.e., temperature, precipitation, precipitation seasonality, lightnings, TCD, settlement time, water distance, and agriculture distance. To define the optimal number of neighbours for the local relationships, we applied the golden search method, which aims to minimize the AICc values and accounts for the spatial distribution of the neighbours. To validate the model's predictive performance, we trained the model on 70% of the locations (i.e., 544 fire locations and 1215 non-fire locations) and compared the predicted probabilites to the classification of the remaining 30% of the mapped locations (see Fig. S1b for spatial distribution of training and validation locations).

For predicting the forest fire hazard across the entire study area, the GWR required polygons or points as input data. We therefore created a set of point data ($n = 155{,}285$) that were regularly distributed across the forest in the study area with a distance of 1 km between the points. For each point, we extracted the values of the predictor variables that were mapped at a spatial resolution of 100 m, matching the variables of the training data. Finally, we applied the GWR model to map the current forest fire hazard across the ASP.

2.6. Assessing the effects of climate change

To predict future forest fire hazard under three different Shared Socioeconomic Pathways (SSPs), we applied the same GWR model as for mapping the current forest fire hazard, replacing current climate data with projections for the period 2070–2100. The other predictor variables remained unchanged to disentangle the effects of anthropogenic-induced climate change on the forest fire hazard. We used the mean annual temperature of the Coupled Model Intercomparison Project phase 6 (CMIP6) climate models from the CHELSA V2.1 data repository (Karger et al., 2017, 2021), including five General Circulation Models (GCMs) for three different SSPs. The three scenarios SSP126, SSP370, and SSP585 represent the lowest to highest values of anthropogenic radiative forcing, global CO_2 concentration, and global CO_2 emissions combined with different socioeconomic developments (O'Neill et al., 2016). The SSP126 is a the most optimistic scenario compatible with the 2 °C target under the assumption of adopting sustainable pathways and implementing climate protection measures. The SSP370 is characterized by regional rivalry, representing the upper-middle range of impacts, while the SSP585 with an additional radiative forcing of 8.5 W/m^2 by the year 2100 represents the worst-case scenario based on an intensified exploitation of fossil fuel resources.

For each SSP, we built a multi-model ensemble by calculating average values across all five GCMs to reduce the uncertainty related to individual climate models (Lehner et al., 2020). Finally, we re-ran the GWR to map the future forest fire hazard under the three SSPs, using the same model settings as for mapping the forest fire hazard under current conditions. To assess the effects of climate change, we compared the outputs of the GWR under current climate conditions with those of the three SSPs. We overlaid the scenario maps with the current forest fire hazard map to map and analyse future potential changes in spatial patterns of the probability of forest fire hazard by calculating the differences in predicted values derived from GWR binary logistic model, i.e., 0 (probability <0.5) and 1 (probability ≥ 0.5). To synthesize the spatial results, we derived mean values across the entire ASP, at the national level, and for elevation zones. Group differences were tested applying one-way ANOVA with the Bonferroni post hoc test in SPSS Statistics software V29.0.1.0 (IBM Cooperation).

3. Results

3.1. Factors influencing the forest fire hazard

Between 2000 and 2023, most of the 772 forest fire events happened in the southern part of the ASP (Fig. 1). The events were concentrated on the late winter/early spring season between January and April (52.3 %; Fig. S3), a second peak occurred during summer in July and August (29.9 %). Least fire events occurred in May, November, and December (2.8%).

The binary logistic regression model explained 57% of the variance in the occurrence of forest fires in the ASP by eight variables (adjusted $R^2 = 0.57$, $p < 0.001$). The model coefficients suggest that the fire hazard mostly depended on the frequency of lightnings and the temperature. In addition, greater Euclidean distance from settlements, water, and agricultural areas, as well as higher precipitation sums and seasonality increased the fire hazard, whereas higher TCD reduced the fire hazard (Table 2). However, the residuals of the regression showed strong spatial autocorrelation (Moran's I: 0.39, z-score: 131.7, p-value: <0.001). Thus, a GWR model was applied to describe local relationships. With the GWR model, the adjusted R^2 increased from the global to the

Table 2
Result of the binary logistic regression model. Significant variables ($p < 0.05$) are indicated in bold.

Variable	Coefficient	Std. Error	z-Statistic	Probability	Odds Ratio	Wald's Low (95%)	Wald's High (95%)	VIF
Intercept	−10.4591	0.6703	−15.6039	<0.0001	0.0000	0.0000	0.0001	–
Aspect	0.0000	0.0012	−0.0301	0.9760	1.0000	0.9975	1.0024	1.0076
Temperature	**0.6046**	**0.0344**	**17.5626**	**<0.0001**	**1.8306**	**1.7111**	**1.9584**	**1.6488**
Precipitation	**0.0008**	**0.0003**	**3.0317**	**0.0024**	**1.0008**	**1.0003**	**1.0013**	**1.9932**
Precip. seasonality	**0.0801**	**0.0093**	**8.6619**	**<0.0001**	**1.0834**	**1.0640**	**1.1033**	**1.5786**
TCD	**−0.0418**	**0.0037**	**−11.3395**	**<0.0001**	**0.9591**	**0.9522**	**0.9661**	**1.2030**
Lightning	**0.8471**	**0.1012**	**8.3722**	**<0.0001**	**2.3329**	**1.9132**	**2.8446**	**1.5156**
Agriculture distance	**0.0007**	**0.0001**	**7.8643**	**<0.0001**	**1.0007**	**1.0005**	**1.0009**	**1.3373**
River distance	−0.0001	0.0001	−0.8225	0.4108	0.9999	0.9998	1.0001	1.1517
Water distance	**0.0000**	**0.0000**	**2.6404**	**0.0083**	**1.0000**	**1.0000**	**1.0000**	**1.1638**
Settlement time	**0.0371**	**0.0055**	**6.7307**	**<0.0001**	**1.0378**	**1.0266**	**1.0490**	**1.4098**
Population 1 km	0.0000	0.0000	1.0986	0.2719	1.0000	1.0000	1.0000	1.4932
Population 5 km	0.0000	0.0000	1.3299	0.1835	1.0000	1.0000	1.0000	1.5155
Visitation Flickr	0.0073	0.0053	1.3821	0.1669	1.0073	0.9969	1.0178	1.3657
Vistitation GBIF	−0.0011	0.0032	−0.3590	0.7196	0.9989	0.9926	1.0051	1.2373

local model from 0.57 to 0.90 and the AIC (Akaike Information Criterion) value decreased from 1351 to 521, respectively. Although the residuals showed less spatial autocorrelation, spatial dependencies in the residuals could not be completely removed with the GWR (Moran's I: 0.06; z-score: 21.9; p-value: <0.001). The validation confirmed the model's capacity to predict the forest fire hazard, as 96.4% of 752 locations, which were not used for training the model, were correctly predicted (93.4% for fire locations, 97.7% for non-fire locations; see Fig. S4 for location of correctly/incorrectly predicted locations and deviation residuals).

The influence of the local coefficients varied across the study area for most variables, mostly between eastern and western and to a less extent between northern and southern parts of the ASP (Fig. 3). For example, precipitation was positively related to forest fires in Eastern Austria, while having negative influence in the central part of the ASP, especially in Switzerland. Temperature, distance from water, and time from settlement had higher positive influence on forest fire hazard in the eastern part of the ASP than in the western parts, whereas it was the contrary for precipitation seasonality and lightning. The predominantly negative influence of TCD turned into a positive relationship in the southwestern Alps, while it was the contrary for distance from water. The distance from agricultural areas had varying influence mostly in the lowlands.

3.2. Effects of climate change on the forest fire hazard

The current forest fire hazard clearly varied across the study area (Fig. 4) due to differences in climate- and vegetation-related characteristics as well as anthropogenic variables (Table S2, Fig. S2). High probability of forest fire hazard occurred in the southwestern part of the ASP in France, as well as in Italy, along the hillslopes and southern parts

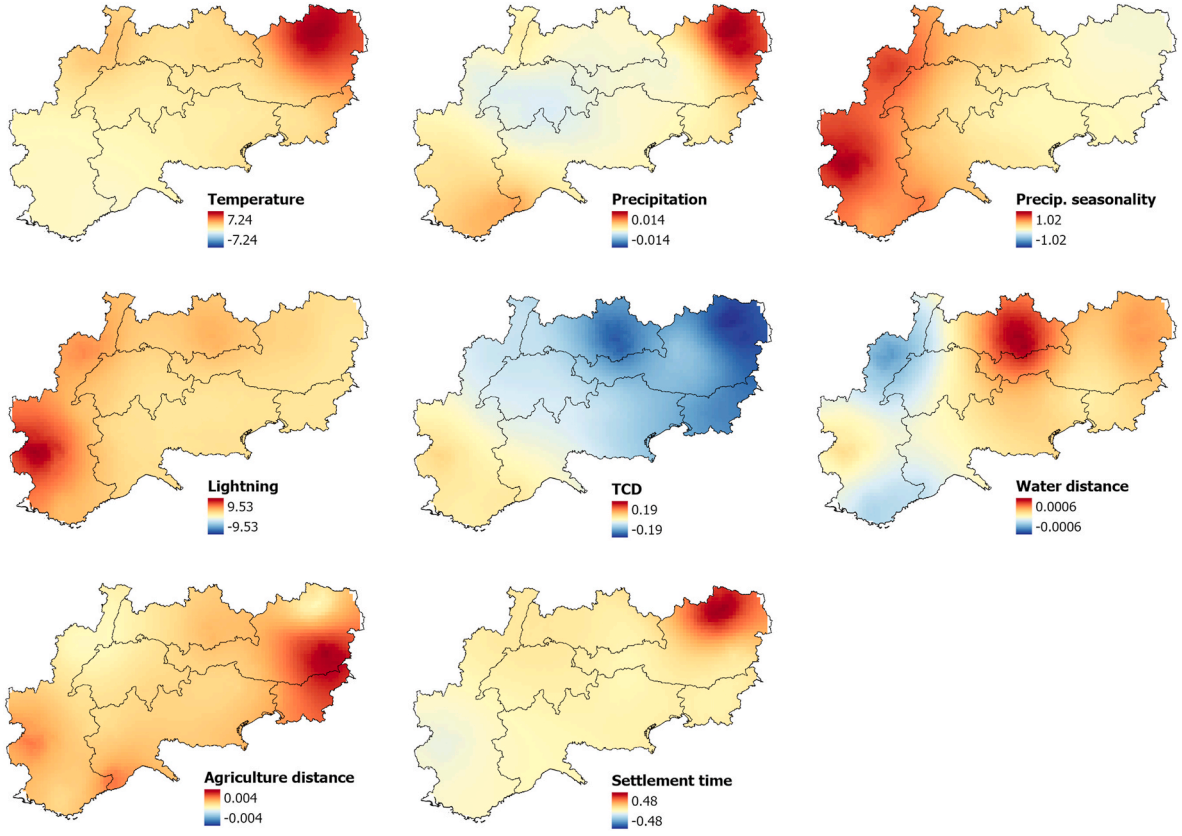


Fig. 3. Spatial variation in the estimated GWR coefficients. Please note that the value range differs across the coefficients to clearly depict spatial differences.

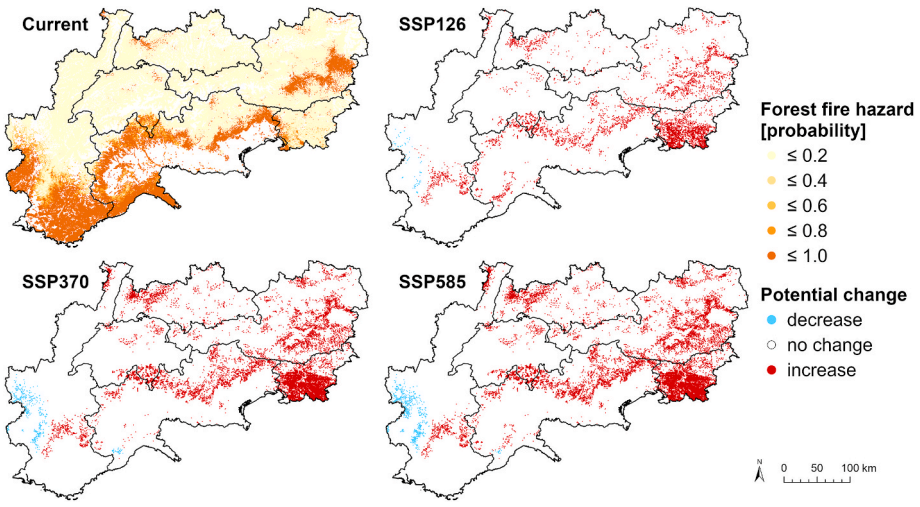


Fig. 4. Current probability of forest fire hazard in the ASP (upper left). Non-forested areas are depicted in white. Potential changes in future probability of forest fire hazard under the three SSPs (upper right, lower left and right). Changes were derived by calculating the difference between current and future predicted values from the GWR binary logistic model, including 0 = low forest fire hazard (probability <0.5) and 1 = high forest fire hazard (probability ≥0.5), i.e., decrease <0, no change = 0, and increase >0. For maps showing the future probability of forest fire hazard, see Fig. S5.

of the European Alps and the Apennine in Liguria. Furthermore, some low-elevated areas in southeastern Austria had high probability of forest fire hazard. In contrast, the high-elevated areas of the European Alps as well as the northern and western lowlands had mostly the low probability of forest fire hazard. The predicted spatial patterns of current forest fire hazard were closely aligned with the actual occurrence of past fire events (Fig. S4), except for some parts in southeastern Austria where the model overestimated the forest fire hazard.

Climate change generally increased the probability of forest fire hazard under the three SSPs in most parts of the study area (Fig. 4, Fig. S5). It significantly increased from currently 0.30 to 0.36 (15.5%) under the SSP126, to 0.40 (30.6%) under the SSP370, and to 0.41 (35.4%) under the SSP585 (Table S3, Table S4). Most increases occurred in the eastern parts of the ASP, particularly in Slovenia, Italy, and Austria, while least increases were found in the southwestern parts in France that already had a high forest fire hazard (Fig. 4). The increase in forest fire hazard significantly ($p < 0.001$) differed across elevation zones as well as in each country (Table S4). The current forest fire hazard was generally higher in the colline zone than at higher elevations (Fig. 5; Table S3), but future increases in the average forest fire hazard were about three times higher in the subalpine zone (ranging between +24.6% under the SSP126 and + 61.6% under the SSP585) compared to the colline zone (ranging between +10.5% under the SSP126 and + 19.1% under the SSP585). At the national level, highest increases occurred in Slovenia (316.5% under the SSP585) and Germany (246.6% under the SSP585), followed by Switzerland (77.5% under the SSP585), and Austria (71.8% under the SSP585), while France and Italy already had high forest fire hazard (0.45 and 0.54, respectively), which increased to a smaller extent under the projected climate changes (2.1% (not significant) and 20.8%, respectively, under the SSP585) (Fig. 5).

4. Discussion

4.1. Factors influencing forest fires in the European Alps

Relating a comprehensive set of environmental and anthropogenic variables to the occurrence of past forest fire events provided insights into the underlying mechanisms in the ASP. The results of the regression models indicate that the occurrence of forest fires was mainly related to high lightning frequencies, high mean annual temperatures, and greater precipitation seasonality, which generally aligns with many other studies (e.g., Babu et al., 2023; Castedo-Dorado et al., 2011; Müller et al., 2020; Sevinc et al., 2020). Spatial patterns of temperature and precipitation data is closely related to topography (Pepin et al., 2022), whereas precipitation seasonality is more influenced by the climate zone, i.e., regions located in mediterranean or continental climate zones

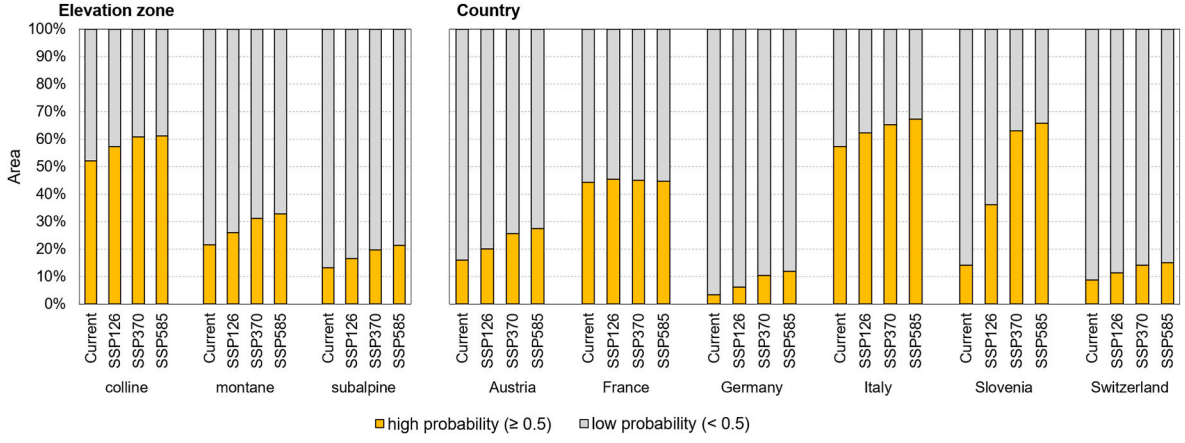


Fig. 5. Potential changes in area of high (≥0.5) and low (<0.5) probability of forest fire hazard under the three SSPs compared to the current situation, distinguished by elevation zone (left) and country (right).

exhibit higher differences in precipitation over the year than regions influenced by the atlantic ocean (see also Fig. S2). While climate data are commonly used by studies predicting the occurrence of forest fires, lightnings are less often considered, although they are the most important natural cause for forest fires (Sevinc et al., 2020). Currently, lightning densities are higher in lower elevated areas, particularly in Italy, Slovenia and eastern Austria (Fig. S2), but climate change is likely to increase lightning activities, which, especially when coinciding with drought conditions, may have considerable impacts in the frequency of forest fires at higher elevations, leading to greater burnt areas and costs (Conedera et al., 2006). Spatial differences in the TCD also influenced the forest fire hazard in our study, which can be explained by differences in tree species composition across the study areas and their specific characteristics in terms of flammability (De Rigo et al., 2016; Popović et al., 2021). Unlike other studies (Babu et al., 2023; Müller et al., 2020; Sevinc et al., 2020), anthropogenic variables played a less significant role in our study region, although most forest fires in the European Alps are caused by human activities (Wastl et al., 2013). This may be primarily explained by the fact that many human-caused forest fires are often quickly detected and therefore can be kept small due to an efficient network of fire brigades (Müller et al., 2020), and thus, were not included in the EFFIS database. This may also a reason for the overestimation of current forest fire hazard in some areas such as in eastern Austria. Nevertheless, some patterns emerged from the GWR, indicating the varying and partly contrasting influence of distance to water, agricultural areas, as well as settlements. These patterns mostly reflect differences in the spatial distribution of these variables (Fig. S2). For example, cultivation of agricultural products is more important in lowland areas in the southern Alps than northern and high-elevated regions that are dominated by livestock farming and related grassland areas (Tasser et al., 2024).

4.2. Effects of climate change on the forest fire hazard and management implications

We found an overall increase in the probability of forest fire hazard across the study area due to climate change, confirming the findings of previous studies in the European Alps (e.g., Cane et al., 2013; Miller et al., 2024; Schumacher and Bugmann, 2006; Stritih et al., 2024; Varela et al., 2019) and at global level (Dupuy et al., 2020; Jones et al., 2022). Similar to other studies (Curasi et al., 2024; Schwörer et al., 2024; Zhang et al., 2024), predicted changes were most evident under the intermediate scenario (SSP370) and the worst-case scenario (SSP585), but the increase in the probability of forest fire hazard was already significant under the most positive scenario (SSP126) in our study area. However, our maps indicate an uneven spatial increase across the European Alps, suggesting that areas that have been less exposed to forest fires in the past will increase in probability of being affected by forest fire hazard in the future, while the impacts were less pronounced in those regions that already had a high forest fire hazard. The spatial expansion of the forest fire hazard was only related on mean climatic changes, but it can be expected that shifts in precipitation patterns will induce longer and more frequent drought periods, also in currently more humid regions (Bowman et al., 2020; Kotlarski et al., 2023; Miller et al., 2024; Wastl et al., 2013). Such developments are particularly critical, as the regions with the highest increase in the probability of forest fire hazard have a high share of forest, which are often also characterized by a huge amount of fuels accumulated (Resco de Dios et al., 2021; Rigo et al., 2017).

These findings have great implications for the adaption of the forest fire management in the different countries. While fire-smart forest management is already in the focus of managers and decision-makers in the fire-prone mediterranean areas (Fernandes, 2013; Pais et al., 2020; Sil et al., 2024), there will be a stronger need for the development of pre-fire management strategies in the currently less affected areas of the European Alps to avoid large forest fires (Müller et al., 2020). This will be particularly important in dynamic landscapes subject to the abandonment of mountain grassland with subsequent shrub and tree encroachment, leading to and increasing forest cover (Bebi et al., 2017; Tasser et al., 2024), connectivity, and fuel build-ups (Valese et al., 2014). One potential measure may be the promotion of tree species that are less flammable due to their bark thickness, leaf characteristics, and litter production (Popović et al., 2021). Second, the fire behaviour can be influenced by establishing spatial heterogeneity in stand structure, which helps to limit the severity and spread of fires (Stephens et al., 2010). In addition, a higher diversity at the stand and landscape level improves the resilience of forests against disturbances (Seidl et al., 2018). If carefully applied and monitored, prescribing adequate fire regimes to reduce the fuel load may be an effective strategy to preserve biodiversity and cultural landscapes (Fernandes et al., 2013; Valese et al., 2014).

4.3. Limitations and future directions

Our results highlight potential impacts of climate change on the forest fire hazard across the ASP, but there are also several limitations. First, the EFFIS database, providing a harmonised dataset for Europe, has the disadvantage that it is mapped with MODIS satellite imagery at 250 m ground spatial resolution and burnt areas are usually greater than 30 ha, while it does not comprise forest fires that could be extinguished before burning a greater area. For our study, this may have led to a restricted and biased sample of past occurrences of forest fires. National databases may offer more complete data, e.g., for Austria (https://fire.boku.ac.at/firedb/en/) and France (https://bdiff.agriculture.gouv.fr/incendies), but it was not possible in this study to build a harmonised dataset for study area extending across different countries. Furthermore, the strong seasonality of the fire events in the study area suggests analysing the forest fire hazard for specific periods, which would be possible with a greater sample of fire events. In this study, to assess long-term impacts of climate change, seasonality was represented by the variable precipitation seasonality, while we also included annual mean temperature and precipitation data, which does not sufficiently capture seasonal patterns. However, it also needs to be considered that climate models generally predict a seasonal shift of precipitation amounts, i.e., precipitation increases in winter and decreases in summer (Kotlarski et al., 2023), which may alter current fire regimes in the European Alps (Valese et al., 2014).

A second limitation was caused by the lack of currently available data, preventing to include further predictor variables. For example, studies suggest that variables related to the forest management such as stand age, forest structure, as well as fuel properties of understory, litter, and dead wood are key factors for the ignition and behaviour of forest fires (Hanes et al., 2021; Morgan Varner et al., 2015; Rigo et al., 2017; Sánchez-Pinillos et al., 2021). Thus, it would be central to consider such variables in future studies, at least, for smaller study areas. Furthermore, the Normalized Difference Vegetation Index (NDVI) can indicate vegetation health and be used as proxy for the availability of fuel (Chuvieco et al., 2004; Ljubomir et al., 2019), which has been successfully improved the modelling performance in studies focusing on the current forest fire hazard (e.g., Babu et al., 2023; Janizadeh et al., 2023; Parvar et al., 2024). In this study, we decided to not use this indicator due to its dependence on phenological patterns that vary with elevation and vegetation type as well as its susceptibility to short term changes (Spadoni et al., 2020), in addition to issues in long-term predictions of the NDVI (Guo et al., 2024). Nevertheless, future studies should consider including it, if they focus on forest fire hazard in specific seasons or for short-term predictions.

Another issue is related to model uncertainties of the long-term climate change projections and, in mountainous areas, additionally to scaling issues due to the complex topography (Kotlarski et al., 2023). Although we addressed this issue by building multi-model ensembles, our results indicating potential future changes still may contain an

elevated level of uncertainty. This is less relevant for the general trends in the level of the forest fire hazard and shifts in spatial patterns, but caution is needed for applying our results at the local level. Furthermore, our study reflects general climatic trends, while potential impacts caused by an increase in frequency and severity of extreme events, such as drought, heat waves, and storms, have not explicitly considered (Hicks et al., 2018; Obojes et al., 2022; Tello-García et al., 2020). However, such events are often favouring forest disturbances such as bark beetle outbreaks and windthrows, increasing the hazard for forest fires due to a higher availability of flammable material (Berčák et al., 2023). Moreover, further drivers such as social changes, economic activities, and new governance strategies that will lead to land-use/cover changes and influence management practices are likely to influence the future forest fire hazard. For example, the ongoing increase in forest cover across the European Alps due to land abandonment and the upshift of the treeline (Tasser et al., 2017, 2024) also extends potential forest fire areas, while changes in tree species composition due to the adaption of current forest management practices towards a climate-smart forestry may reduce the forest fire hazard (Seidl et al., 2018).

Considering these limitations, our geostatistical modelling approach can be easily applied in other regions. It allows the integration of further predictor variables, which may improve the reliability of the results. This is particularly important for predictions at the regional or local level, where fine-scale data, for example, related to the forest management may be available.

5. Conclusions

By mapping potential impacts on the forest fire hazard under three SSPs using a geostatistical modelling approach, our findings contribute to a broader understanding of potential future changes across the European Alps and surrounding lowlands. In general, our results indicate an overall expansion in the probability of forest fire hazard that is already significant under the most positive scenario (SSP126), but the maps also highlight spatial differences. While the impacts are less pronounced in already fire-prone regions, the forest fire hazard will greatly increase in the Northern and Eastern regions. Until now, these regions had been less exposed to forest fires, but climate-related changes will make them to an emerging issue.

Thus, our findings emphasize the urgent need to address these climate-related challenges through the development of fire-smart and climate-smart forest management. At the cross-national level, our findings can support European-wide policies such as the EU Green Deal, including the Forest Strategy, the EU Biodiversity Strategy for 2030, and the new Nature Restoration Law. For example, policy measures should foster an improved management of species diversity in forests, promoting the creation of more heterogenous stands and landscapes that are more resilient against forest disturbances and climate change impacts. At the regional and local level, our findings can support the development of targeted management plans and strengthen the communication with stakeholders in the decision-making process.

Despite providing novel insights into the future forest fire hazard at the cross-regional level, further efforts are needed to overcome current limitations related to data availability and uncertainties of future scenarios. For example, the results could be improved by accounting for climate-related hazards such as drought periods and by integrating more specific data on stand characteristics and forest management. Such improvements are particularly important for predictions at the local level. While this study aimed to disentangle the effect of climate change on the forest fire hazard, further drivers altering the forest expansion and forest structure should be considered in future scenarios.

CRediT authorship contribution statement

Kilian Gerberding: Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Uta Schirpke:** Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the Met Office of the United Kingdom (https://www.metoffice.gov.uk/), which kindly provided the data of the lightning location system ATDnet. The authors thank the Department of Innovation, Research University and Museums of the Autonomous Province of Bozen/Bolzano for covering the Open Access publication costs.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2025.124600.

Data availability

Data will be made available on request.

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